MIL-W-50933 (MU)
9 August 1971
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

WARHEAD SECTION, 318MM ROCKET, HE: M146

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Warhead Section designated as Warhead Section, 318MM Rocket, HE: M146.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1340

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8839602	-	Packing and Marking for Warhead Section 318MM Rocket, HE, M146 or XM185
8836172	-	Warhead Section, 318MM Rocket HE: M146

PUBLICATIONS

ARMY

ET-8836172	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	-	Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify: "49 CFR 0-190 [latest revision[.")

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg.) F-8836172, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Serial Numbers.-Each forward assembly and rear assembly shall be marked with a serial number to assure the same forward assembly will be reassembled to the rear assembly for which it was removed, prior to loading.

3.4 Stability of bag and filler assembly.-The foaming plastic bag shall expand and achieve a uniform firm set when tested as specified in 4.4.3.

3.5 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.4).

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order,

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the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless otherwise disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the cerfitying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a sample of two (2) assemblies to a Government approved facility as designated by the contracting officer for testing in accordance with the provisions of 4.2.2 (See 6.1). All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract. First article inspection shall also apply when a lapse in production exceeds ninety (90) days.

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4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Loaded grenades of one lot interfix number from one manufacturer.
b. Metal parts and metal parts assemblies from one manufacturer and of one lot interfix number.
c. Loaded fuzes of one lot interfix number from one manufacturer.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Bag and filler assembly (See Dwg 8836370).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Cut or tear	Visual	01001
102.	Stitch broken or missing	Visual	01002
103.	Filler missing	Visual	01003
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	01004

4.3.2.2 Warhead Section, 318MM Rocket, HE: M146, prior to installing ring (See Dwg. 8836172).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Bag and filler assembly improperly positioned	Visual	02001

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Minor	AQL 1.50 Percent		
201.	Top side of bag not cemented	Visual	02002
202.	Cover sheet missing	Visual	02003
203.	Evidence of poor workmanship (See 3.5)	Visual	02004

4.3.2.3 Warhead Section, 318MM Rocket, HE: M146, prior to installing forward section (See 8836172).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Bulkhead plate loose	Manual	03001
102.	Retaining ring improperly assembled	Visual	03002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	03003

4.3.2.4 Warhead Section, 318MM Rocket, HE: M146, (See Dwg. 8836172)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Weight	Balance	04001
102.	Concentricity and Perpendicularity of ogive with datum diameter	Gage	04002
103.	Ogive loose	Manual-Visual	04003
Minor	AQL 1.50 Percent		
201.	Nicks, dents, scratches or cuts	Visual	04004
202.	Marking misleading or unidentifiable	Visual	04005
203.	Evidence of poor workmanship (See 3.5)	Visual	04006

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4.3.2.5 Packing and Marking for warhead Section, 318MM Rocket HE, M146 or XM185, prior to securing cover (See Dwg. 8839602)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Warhead Section improperly oriented	Visual	05001
102.	Desiccant insufficient or missing	Visual	05002
103.	Humidity indicator card missing	Visual	05003
104.	Packed fuze missing or improperly secured	Visual-Manual	05004
105.	Forward saddle clamp loose or missing	Visual-Manual	05005
106.	Other hardware loose or missing	Visual-Manual	05006
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	05007

4.3.2.6 Packing and Marking for Warhead Section, 318MM Rocket, HE, M146 or XM185, Sealed (See Dwg. 8839602)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Car seal missing unsealed or improperly attached	Visual-Manual	06001
102.	Latch insecure	Visual-Manual	06002
Minor	AQL 1.00 Percent		
201.	Marking misleading or unidentifiable	Visual	06003
202.	Evidence of poor workmanship (See 3.5)	Visual	06004

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4.3.3 Testing

4.3.3.1 Lateral center of gravity (See Dwg 8836172) Major Defect, Code No. 07001.-Each warhead section shall be subjected to this test. If any warhead section fails to comply with the applicable requirement, the warhead section shall be classed defective and removed from the lots. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Longitudinal center of gravity (See Dwg 8836172) Major defect, Code No. 08001.-Each warhead section shall be subjected to this test. If any warhead section fails to comply with the applicable requirement, the warhead section shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.3 Stability of bag and filler assembly (See 3.4 and Dwg. 8836370), Major defect, Code No. 09001.-One (1) bag and filler assembly shall be selected from each batch of twenty-five (25) or less, prepared and subjected to this test. The filler for each batch shall be homogeneous and mixed at one time. Batches over twenty-five (25) shall be qualified by selecting one sample from each twenty-five (25) or portion thereof produced. If the foaming plastic fails to achieve requirements specified in 3.4, the batch represented by the failing sample shall be rejected and a sample from the next batch stored at -80°F shall be similarily tested. The test shall be performed as specified in 4.4.3.

4.3.3.4 Pressure retention test (See Dwg. 8836172) Major defect, Code No. 10001.-Each container, with the M146 Warhead Section in position, shall be subjected to this test. If any container fails to comply with the applicable requirements, it shall be classed defective and removed from the lot. Test shall be performed as specified in 4.4.4.

4.3.3.5 Check test for deterioration of bag and filler assembly, Major defect, Code No. 11001.-If a batch, or portion thereof, has been stored for more than two (2) months, the test described in 4.4.3 shall be made prior to use in assembly. Sampling as in 4.3.3.3. If any bag and filler assembly fails to comply with requirements specified in 3.4, the lot represented by the sample tested shall not be used in production.

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4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8836172 and Equipment Lists (EL's) referenced theron identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Lateral center of gravity.-The loaded warhead section shall be mounted in a fixture listed on the Equipment list and tested for conformance to drawing requirements. This is a non-destructive test. Assemblies so tested shall be returned to the lot.

4.4.2 Longitudinal Center of gravity.-The loaded warhead section shall be mounted in a fixture listed on the Equipment list and tested for conformance to drawing requirements. This is a non-destructive test. Assemblies so tested shall be returned to the lot.

4.4.3 Stability of bag and filler assembly.-The sample selected shall be allowed to stand at room temperature (70-90°F) for not less than four (4) hours. It shall then be examined visually. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.4 Pressure retention test.-Immediately prior to shipment the container, with the warhead section in position shall be tested for pressure retention as specified on drawing 8839602. The pressure shall be released upon completion of the test.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and packing

5.1.1 Level A.-The warhead section shall be preserved, packaged and packed in accordance with Dwg. 8839602.

5.2 Marking.-Marking shall be in accordance with Dwg. 8839602 and Code of Federal Regulations, Title 49, Parts 171-179.

5.3 Shipping.-When assemblies from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as

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"Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legi-

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bility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.5 Supersession Data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-50 Rev. 1 dated 23 December 1963.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1340-A

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